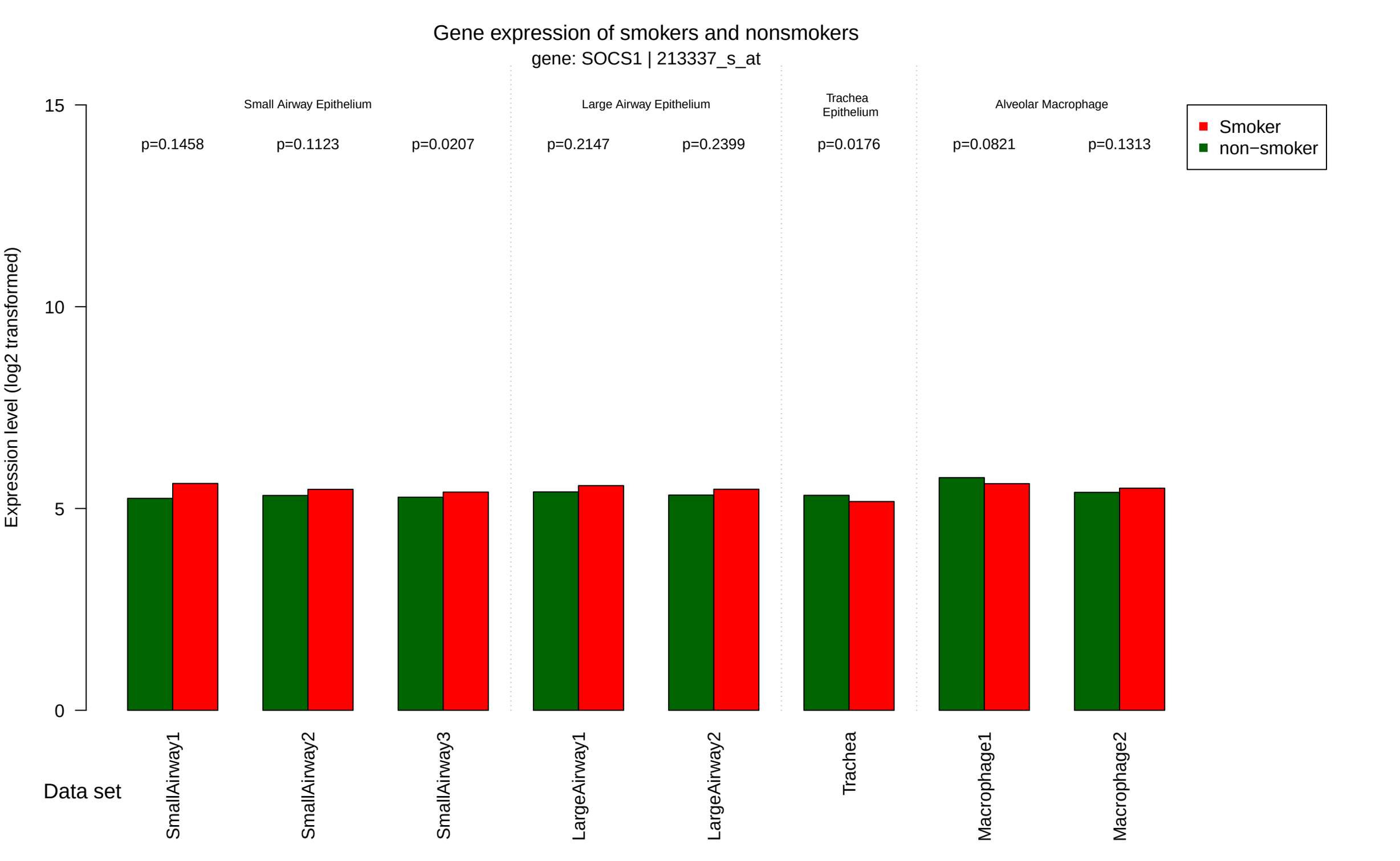
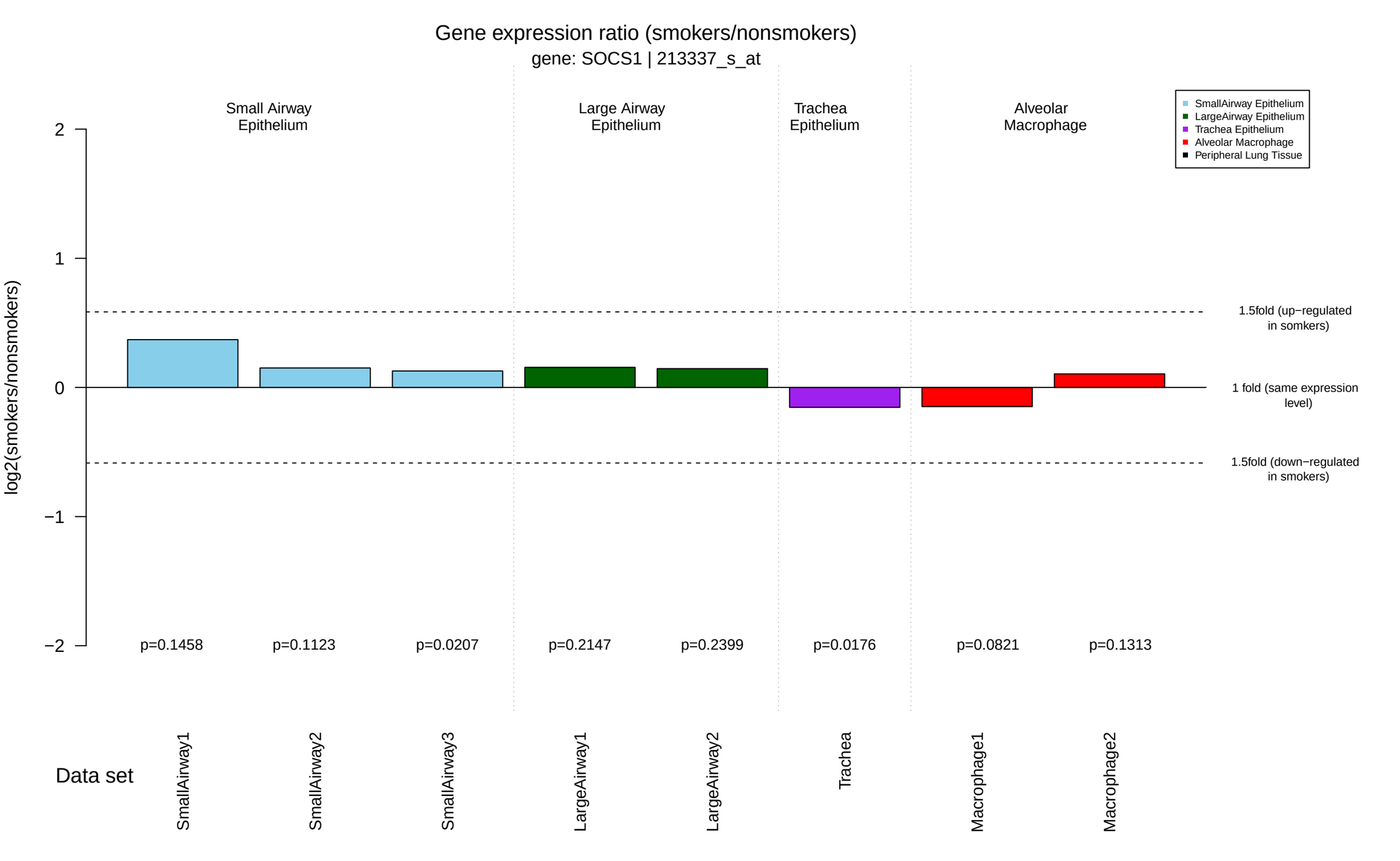
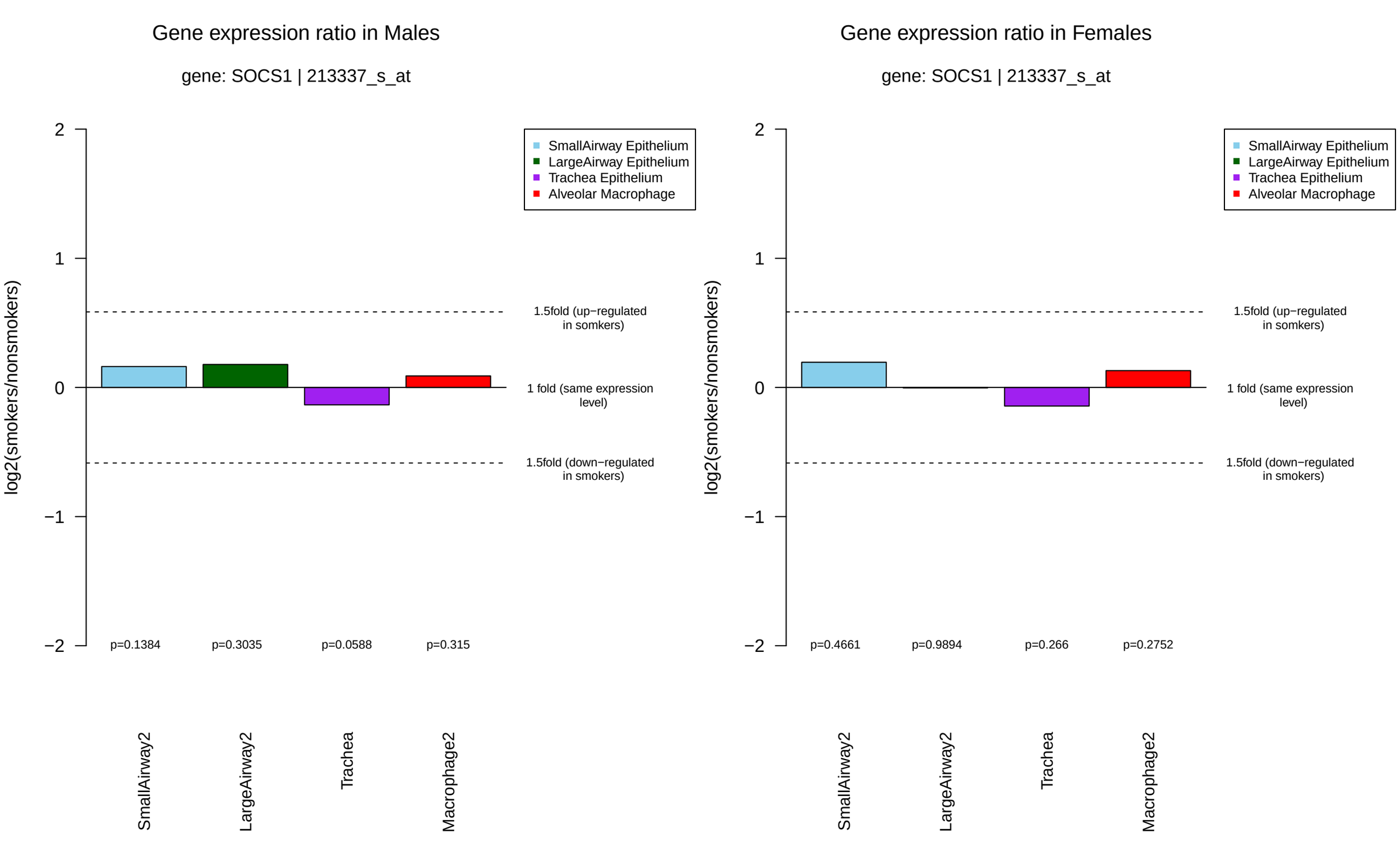
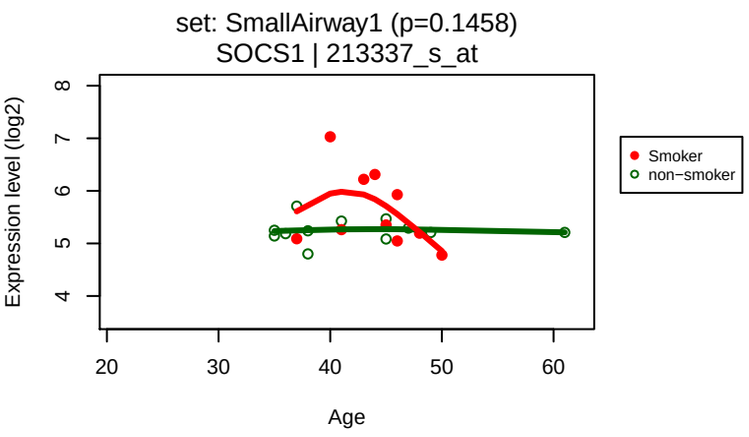
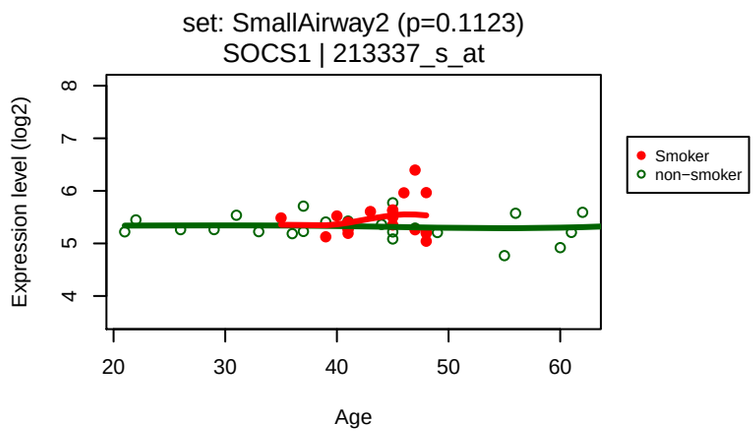
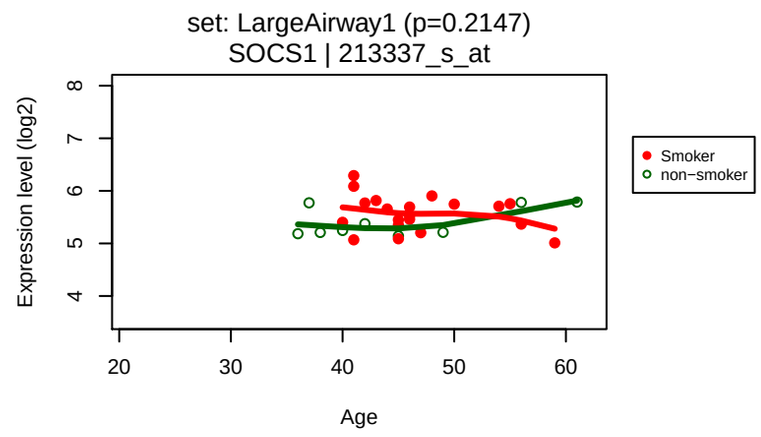

Gene expression of smokers and nonsmokers
gene: SOCS1 | 213337_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1458
p=0.1123
p=0.0207
p=0.2147
p=0.2399
p=0.0176
p=0.0821
p=0.1313
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

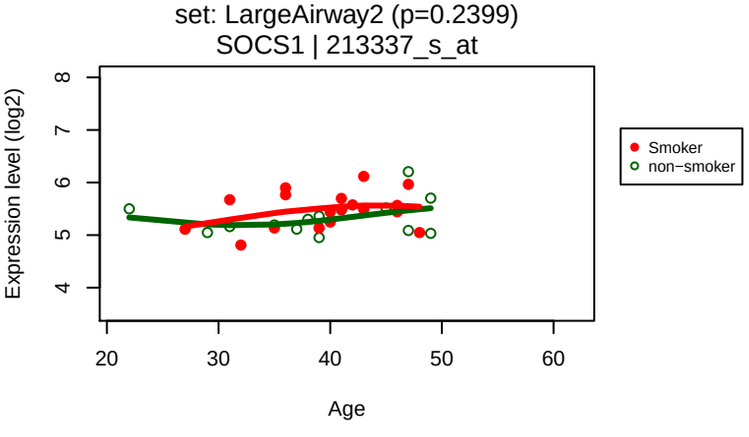

Gene expression ratio (smokers/nonsmokers)
gene: SOCS1 | 213337_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1458
p=0.1123
p=0.0207
p=0.2147
p=0.2399
p=0.0176
p=0.0821
p=0.1313
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

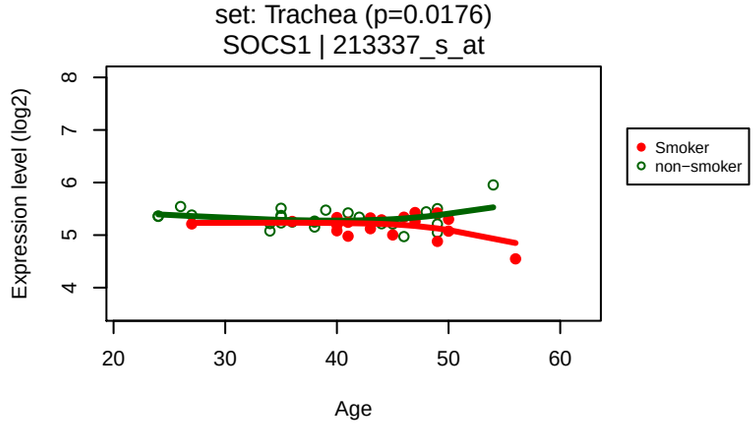

Gene expression ratio in Males
gene: SOCS1 | 213337_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1384
p=0.3035
p=0.0588
p=0.315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS1 | 213337_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4661
p=0.9894
p=0.266
p=0.2752
SmallAirway2
LargeAirway2
Trachea
Macrophage2

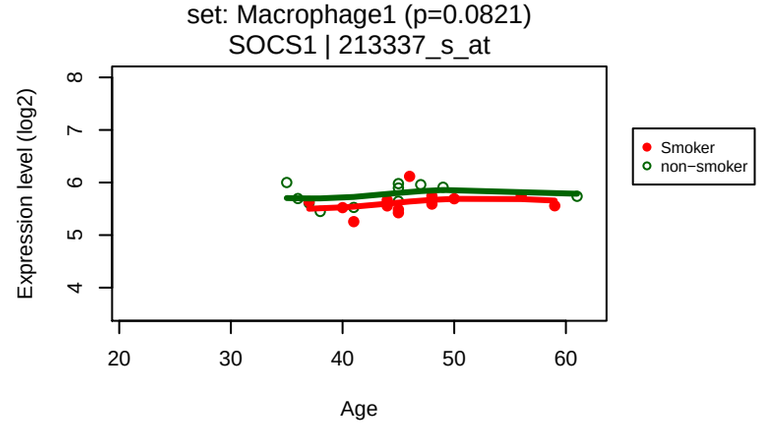

set: SmallAirway1 (p=0.1458)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker

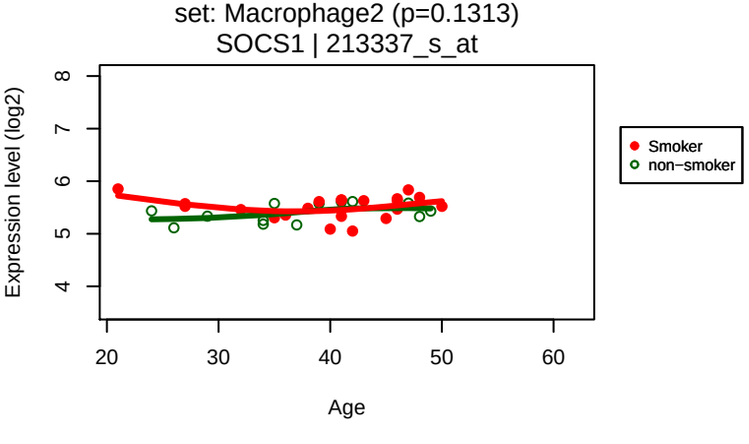

set: SmallAirway2 (p=0.1123)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker

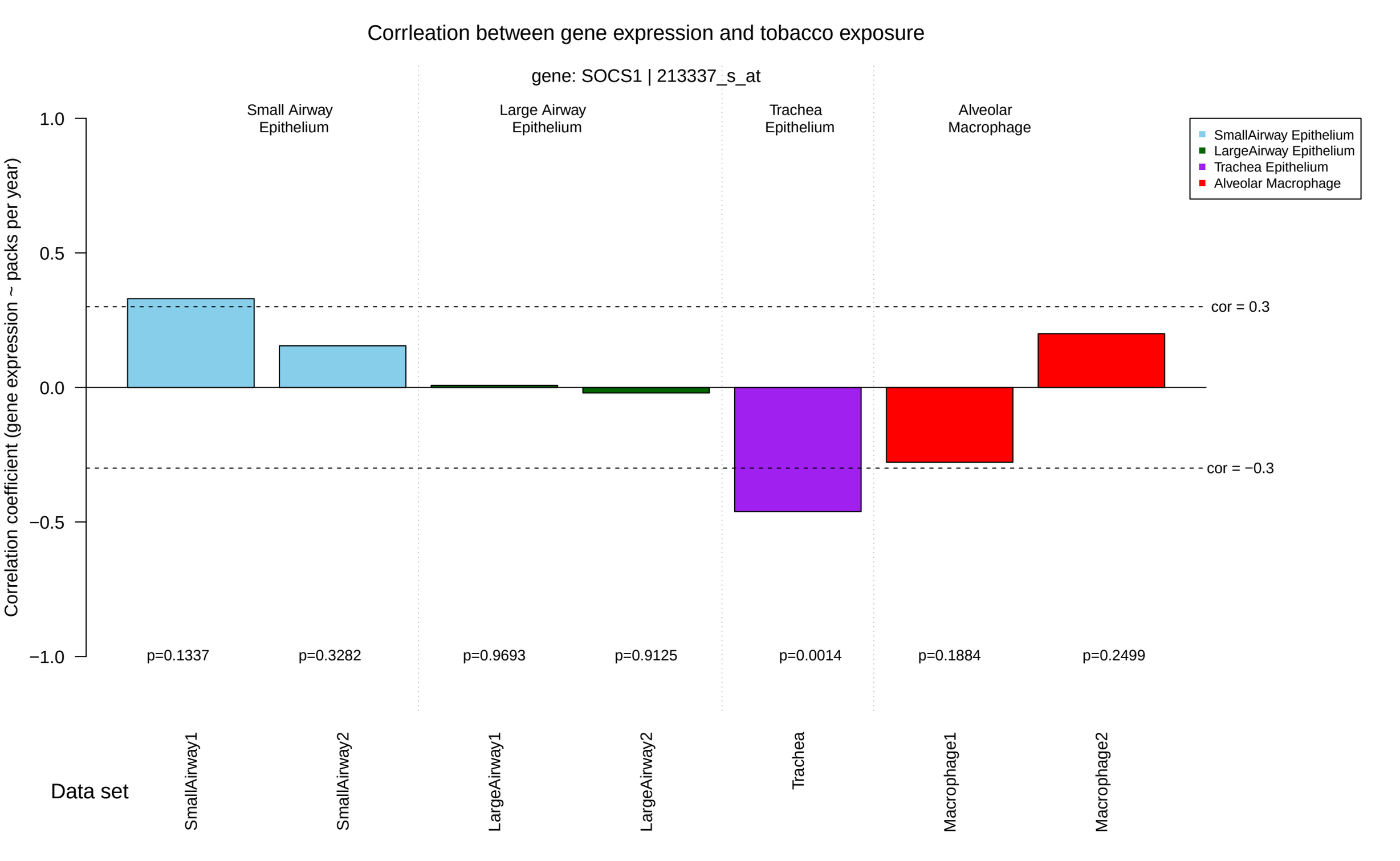

set: LargeAirway1 (p=0.2147)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2399)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0176)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0821)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1313)
SOCS1 | 213337_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS1 | 213337_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1337
p=0.3282
p=0.9693
p=0.9125
p=0.0014
p=0.1884
p=0.2499
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2